

Title: "Falsifiable Predictions of the DK-RD2 (Duo-Kinetic Relativistic Dynamics 2) Framework: Observational Tests Beyond Λ CDM"

Jazmín Olvera Zacarias

* ORCID: 0009-0007-1771-4076 E-mail: Jaolveraz@ipn.mx

Gabriel Martín del Campo Flores

* ORCID: 0009-0009-7106-9519 E-mail: gabemdelc@gmail.com

Abstract

The standard Λ CDM cosmological model successfully reproduces a broad range of observations but relies on a cosmological constant and on dark-sector components whose physical origin remains unknown. This motivates the exploration of alternative frameworks capable of producing falsifiable predictions that distinguish between competing explanations of cosmic acceleration, structure formation, and gravitational phenomena across multiple astrophysical scales.

In this work, we present a set of observational tests derived from the DK-RD2 (Duo-Kinetic Relativistic Dynamics 2) framework, in which the effective gravitational coupling is described by

$$G_{ab}(T, v) \approx G_0 \cdot \left(1 + \alpha_{DK} \frac{v^2}{c^2} \cdot \frac{T_0}{T} \right)$$

where the gravitational interaction evolves according to the thermodynamic state of the system and its characteristic relativistic dynamics.

Within this formulation, the observed gravitational behavior emerges from thermodynamic–relativistic processes rather than from an explicit dark-energy sector. These predictions can be tested using existing and forthcoming datasets from DESI, Euclid, the Vera C. Rubin Observatory, the Nancy Grace Roman Space Telescope, gravitational-lensing surveys, compact-halo observations, and independent measurements of the Hubble constant.

Eight primary observational signatures are identified, including statistical calibration and likelihood constraints on the projection parameter α_{DK} , thermodynamic–relativistic coupling reconstruction, effective dark-energy diagnostics, large-scale structure growth, compact-halo amplification effects, Hubble-constant projection signatures, and observational discriminators capable of distinguishing DK-RD2 from alternative explanations of recent DESI observations.

In addition, the framework is compared with alternative interpretations of recent DESI observations, including evolving dark-energy models, Hot New Early Dark Energy (Hot NEDE[11]), and Dark Acoustic Oscillation scenarios. This comparison highlights the distinct observational signatures predicted by DK-RD2 and provides a direct pathway for experimental discrimination among competing cosmological frameworks. Together, these tests provide a unified observational framework spanning cosmological expansion, structure formation, gravitational lensing, compact gravitational systems, and the Hubble-tension problem through a modular and fully reproducible open-source Python pipeline.

The objective of this work is not to establish the validity of DK-RD2 a priori, but to provide a clear observational roadmap capable of confirming or falsifying the framework. By emphasizing predictive power, multi-scale observational discrimination, comparison with competing cosmological scenarios, and independent reproducibility rather than cosmological fitting alone, the proposed tests provide a direct methodology for evaluating whether cosmic acceleration, structure formation, compact gravitational phenomena, and the observed Hubble tension can emerge from thermodynamic–relativistic processes without requiring an explicit dark sector or cosmological constant.

Rather than further optimizing cosmological fits, the goal is to identify a set of specific observational signatures capable of distinguishing DK-RD2 from Λ CDM and from alternative explanations of recent cosmological observations.

Keywords: DK-RD2; Duo-Kinetic Relativistic Dynamics 2; Emergent Gravity; Thermodynamic Gravity; Modified Gravity; Observational Cosmology; Cosmic Acceleration; Hubble Tension; Baryon Acoustic Oscillations; Gravitational Lensing; Structure Growth; DESI; Model Discrimination; Cosmological Tensions; Falsifiable Cosmology.

1. Introduction

Modern cosmology stands at a conceptual and observational crossroads. The standard Λ CDM model successfully reproduces[7, 8] a broad range of high-precision observations, including the Cosmic Microwave Background (CMB), Type Ia supernovae[4,5], Baryon Acoustic Oscillations (BAO), and large-scale structure formation. However, this agreement relies on the introduction of a cosmological constant[6] together with dark matter and dark energy components, which collectively account for approximately 95% of the total energy content of the Universe while their fundamental physical nature remains unknown.

Recent observational programs, including the Dark Energy Spectroscopic Instrument (DESI)[14], have reported indications that the effective dark-energy equation of state may evolve with redshift rather than remain strictly constant. At the same time, independent determinations of the Hubble constant continue to exhibit a statistically significant discrepancy between early- and late-universe measurements, commonly referred to as the Hubble tension. These observations have stimulated the development of numerous alternative cosmological scenarios, including evolving dark-energy models, Early Dark Energy (EDE), Hot New Early Dark Energy (Hot NEDE[11]), Dark Acoustic Oscillation (DAO) models, modified-gravity frameworks, and related approaches.

The existence of multiple competing explanations highlights a central challenge of modern cosmology: reproducing existing observations is no longer sufficient. Distinguishing between alternative physical mechanisms requires models capable of gen-

erating clear, testable, and potentially falsifiable predictions across multiple independent observables.

In parallel, several theoretical approaches have explored gravity as a potentially emergent phenomenon. Thermodynamic, entropic, and geometric formulations suggest that gravitational dynamics may arise from deeper microscopic processes[9, 10], providing alternative pathways for interpreting cosmological observations without introducing additional dark components. Within this broader context, the DK-RD2 (Duo-Kinetic Relativistic Dynamics 2) framework proposes that the effective gravitational interaction depends on the thermodynamic and relativistic state of the physical system.

The central coupling of the framework is described by

$$G_{ab}(T, v) \approx G_0 \cdot \left(1 + \alpha_{DK} \frac{v^2}{c^2} \cdot \frac{T_0}{T} \right)$$

where G_0 is Newton's gravitational constant, T is the local thermodynamic temperature, T_0 is a reference temperature, v represents the characteristic relativistic dynamics of the system, and α_{DK} is an effective geometric projection parameter. Within this formulation, gravitational behavior emerges as a thermodynamic–relativistic phenomenon whose observable effects evolve throughout cosmic history.

For cosmological applications, the thermal evolution of the Universe is approximated

$$T(z) \approx T_0(1 + z)$$

which naturally suppresses the DK correction at high redshift while maximizing its observable contribution near the present cosmological epoch.

Previous DK-RD2 studies have shown that this framework can reproduce a variety of cosmological observables without introducing an explicit dark-energy sector. However, reproducing existing observations is not sufficient to establish the physical relevance of any cosmological model. A scientifically meaningful framework must also generate clear predictions that can be independently tested and potentially falsified by future observations.

The purpose of the present work is therefore not to further optimize cosmological fits, but to identify a set of specific observational signatures that distinguish DK-RD2 from Λ CDM and from other competing explanations of recent cosmological observations. Particular emphasis is placed on predictions that can be directly tested using current and forthcoming observational programs, including DESI[14], Euclid, the Vera C. Rubin Observatory, the Nancy Grace Roman Space Telescope, future gravitational-lensing surveys, compact-halo observations, and independent measurements of the Hubble constant.

To achieve this objective, we investigate eight observational signatures designed to test, constrain, and potentially falsify the DK-RD2 framework using reference templates within a modular open-source Python pipeline:

1. Statistical calibration of the projection parameter α_{DK}
2. Likelihood constraints on α_{DK}
3. Thermodynamic–relativistic coupling structure and cosmological projection
4. Effective dark-energy diagnostics reconstructed from the DK-RD2 expansion history
5. Modified growth histories detectable through large-scale structure surveys
6. Localized gravitational amplification effects in compact galactic and sub-galactic systems
7. A thermodynamic projection mechanism capable of generating distinct Hubble-constant reconstructions between early- and late-universe observations
8. Observational discrimination between DK-RD2 and alternative explanations of recent DESI[14] observations

The first six observational signatures probe the framework across cosmological and astrophysical scales through expansion history, lensing geometry, structure growth, and compact gravitational systems. The seventh extends the analysis to the Hubble-tension problem, providing an independent test of whether differences between early- and late-universe measurements may emerge naturally from the thermodynamic evolution of the effective gravitational coupling. The eighth introduces a comparative observational framework designed to distinguish DK-RD2 from competing interpretations of recent DESI[14] observations, including Hot NEDE[11] and Dark Acoustic Oscillation scenarios. Together, these eight observational signatures define an experimental roadmap capable of confirming or falsifying the DK-RD2 framework through direct comparison with current and forthcoming observational data spanning multiple physical scales.

By focusing on predictive power, statistical reproducibility, observational discriminators, and comparison with competing cosmological scenarios rather than model fitting alone, this work aims to provide a transparent and testable framework for evaluating whether cosmic acceleration, structure formation, compact gravitational phenomena, and the observed Hubble tension can emerge from thermodynamic–relativistic processes without requiring an explicit dark sector or cosmological constant[6].

The present analysis should be interpreted as a background-level and phenomenological falsification program. A complete validation of DK-RD2 will require a covariant formulation of the coupling and a first-principles implementation of modified perturbation dynamics in a Boltzmann solver.

2. Physical Basis of the DK-RD2 Predictions

The DK-RD2 (Duo-Kinetic Relativistic Dynamics 2) framework has been presented in previous works as a thermodynamic–relativistic extension of gravitational dynamics in which the effective gravitational interaction depends on the energetic, thermal, and relativistic state of the system rather than remaining strictly constant.

A detailed derivation of the framework, its cosmological reconstruction methodology, and its thermodynamic interpretation have been presented elsewhere[1, 2]. The purpose of the present work is therefore not to rederive the model, but to summarize the fundamental relation from which the observational predictions discussed in the following sections are obtained.

Within DK-RD2, the effective gravitational coupling is written as

$$G_{ab}(T, v) \approx G_0 \cdot \left(1 + \alpha_{DK} \frac{v^2}{c^2} \cdot \frac{T_0}{T} \right) \quad (1)$$

where:

- G_0 is Newton's gravitational constant
- c is the speed of light
- v represents an effective characteristic relativistic velocity associated with the dominant dynamical state of the system
- T is the thermodynamic state of the system
- $T_0 = 2.725$ K is the present-day temperature of the cosmic microwave background (CMB)
- α_{DK} is an effective geometric projection parameter.

In this formulation, gravitational behavior is interpreted as an emergent thermodynamic-relativistic phenomenon whose observable effects evolve as the energetic state of the system changes. The parameter α_{DK} determines the observable fraction of the underlying thermodynamic coupling and plays a central role in the predictions investigated in this work.

For cosmological applications, the thermal evolution of the Universe is approximated by

$$T(z) \approx T_0(1 + z) \quad (2)$$

which naturally suppresses the DK correction at high redshift while progressively increasing its observable contribution as the Universe evolves toward the present cosmological epoch.

The standard Newtonian limit is naturally recovered when

$$\lim_{v \rightarrow 0} G(T, v) = G_0$$

or equivalently when

$$\lim_{\frac{T}{T_0} \rightarrow \infty} G(T, v) = G_0$$

demonstrating that the DK-RD2 coupling preserves the classical gravitational regime under ordinary astrophysical conditions.

Rather than focusing on the theoretical derivation of the framework, the present study concentrates on the observable consequences of this coupling. In particular, we investigate whether the DK-RD2 formulation generates distinct signatures that can be tested through current and future cosmological and astrophysical observations.

The following sections examine eight primary observational signatures:

1. Statistical calibration of the projection parameter α_{DK}
2. Likelihood constraints on α_{DK}
3. Thermodynamic–relativistic coupling structure and cosmological projection
4. Effective dark-energy diagnostics reconstructed from the DK-RD2 expansion history
5. Modified growth histories detectable through large-scale structure surveys
6. Localized gravitational amplification effects in compact galactic and sub-galactic systems
7. A thermodynamic projection mechanism capable of generating distinct Hubble-constant reconstructions between early- and late-universe observations
8. Observational discrimination between DK-RD2 and alternative explanations of recent DESI[14] observations

The first six observational signatures probe the framework across cosmological and astrophysical scales through expansion history, lensing geometry, structure growth, and compact gravitational systems. The seventh extends the analysis to independent determinations of the Hubble constant, providing a direct test of whether differences between early- and late-universe measurements may emerge naturally from the thermodynamic evolution of the effective gravitational coupling.

The eighth prediction introduces a comparative observational framework designed to distinguish DK-RD2 from competing interpretations of recent DESI[14] observations, including Hot New Early Dark Energy (Hot NEDE[11]) and Dark Acoustic Oscillation (DAO) scenarios. Rather than focusing exclusively on agreement with existing datasets, this comparison emphasizes future observational discriminators capable of separating competing physical interpretations of the same cosmological phenomena.

Together, these eight observational signatures define a multi-scale experimental roadmap capable of confirming or falsifying the DK-RD2 framework through direct comparison with current and forthcoming datasets using a fully reproducible open-source Python pipeline.

2.2 Observational Predictions and Falsification Road map

The observational program proposed in this work is designed to evaluate the DK-RD2 framework through multiple independent signatures spanning cosmological and astrophysical scales. Each observable probes a distinct physical regime and provides an opportunity for validation or falsification through direct comparison with current and forthcoming datasets.

Together, the proposed tests investigate whether a single thermodynamic–relativistic gravitational framework can simultaneously account for cosmological expansion, gravitational lensing, structure formation, compact gravitational systems, and the observed discrepancy between early- and late-universe determinations of the Hubble constant.

Beyond testing the internal consistency of the framework, the observational program is also designed to distinguish DK-RD2 from competing cosmological scenarios currently proposed to explain recent observational tensions, including evolving dark-energy models, Hot New Early Dark Energy (Hot NEDE[11]), and Dark Acoustic Oscillation (DAO) frameworks.

The figures presented in the following sections therefore serve two complementary purposes. First, they establish whether DK-RD2 generates statistically meaningful and observationally testable predictions across multiple physical scales. Second, they identify specific observational discriminators capable of separating thermodynamic–relativistic gravitational evolution from alternative explanations of the same cosmological phenomena.

A central aspect of this program is that the proposed observables are not treated as independent phenomenological fits. Instead, they arise from a common thermodynamic–relativistic coupling and therefore generate correlated predictions across expansion history, effective dark-energy diagnostics, structure growth, gravitational lensing, compact gravitational systems, and Hubble-constant determinations. This correlation provides an additional level of falsifiability beyond agreement with any individual dataset.

Taken together, these observational signatures define a multi-scale falsification roadmap spanning cosmic expansion, gravitational lensing, structure formation, compact gravitational systems, Hubble-constant reconstructions, and competing interpretations of recent DESI[14] observations. Future measurements from DESI[14], Euclid, the Vera C. Rubin Observatory, the Nancy Grace Roman Space Telescope, gravitational-lensing surveys, and related observational programs will therefore provide direct opportunities to validate or falsify the framework.

Figure 1 — Statistical calibration of the projection parameter α_{DK}

The first observational prediction concerns the statistical calibration of the DK-RD2 projection parameter α_{DK} . Figure 1 presents the joint determination of α_{DK} and the sound-horizon scale r_d using DESI[14] Baryon Acoustic Oscillation (BAO) measurements together with a Cosmic Chronometer (CC) anchor. In this diagnostic construction, the raw DK-RD2 expansion history is projected through

$$E_{DK,eff}^2(z) = 1 + \alpha_{DK} [E_{DK,raw}^2(z) - 1]$$

while r_d is simultaneously fitted by minimizing the combined BAO + CC likelihood[15, 16]. The best-fit value

$$\alpha_{DK} \approx 0.2974$$

indicates that only a fraction of the raw thermodynamic–relativistic coupling is projected into the large-scale observable expansion.

Figure 01 — DESI Expansion and BAO Calibration of α_{DK}
 $\alpha_{\text{DK}} = 0.2974$, $r_d = 149.99 \text{ Mpc}$

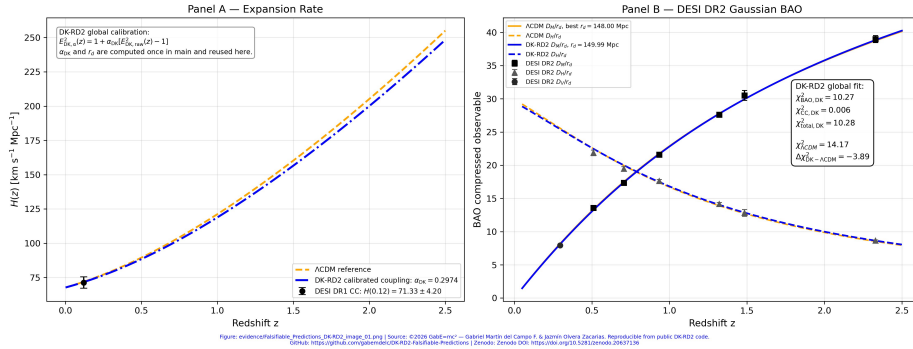


Figure 1 presents the joint calibration of the DK-RD2 projection factor α_{DK} and the sound-horizon scale r_d using DESI BAO measurements and a Cosmic Chronometer anchor.

Importantly, no external dark-energy density parameter is introduced during this calibration. The accelerated-expansion behavior emerges from the thermodynamic evolution of the effective gravitational coupling encoded in the DK-RD2 framework.

The existence of a stable likelihood minimum demonstrates that α_{DK} is observationally constrained rather than arbitrary. Consequently, the parameter becomes a measurable quantity whose value can be refined, tested, or directly falsified by future observations.

This result constitutes the first step of the DK-RD2 observational program, establishing the statistically preferred parameter space from which all subsequent predictions presented in this work are derived.

Figure 2 — Likelihood Constraint on α_{DK}

The second observational prediction examines the statistical likelihood structure of the DK projection factor α_{DK} through the profiled quantity through the profiled quantity $\Delta\chi^2(\alpha_{DK})$, obtained from the combined DESI[14] Baryon Acoustic Oscillation (BAO) and Cosmic Chronometer (CC) datasets. For each value of α_{DK} , the sound-horizon scale r_d is independently optimized by minimizing the joint likelihood

$$\chi^2\text{BAO} + \text{CC}(\alpha_{DK}, r_d)$$

The resulting profile exhibits a clear and stable minimum at

$\alpha_{DK}=0.2974$ with $r_d = 149.99$ Mpc and $\chi^2_{\min}=10.28$

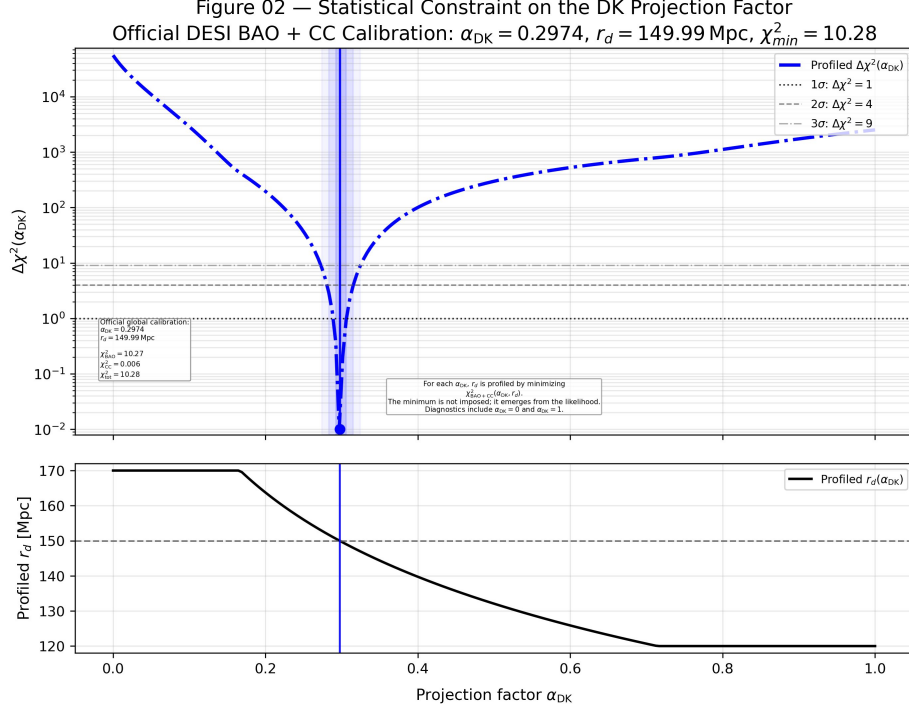


Figure: evidence/Falsifiable_Predictions_DKRD2_Image_02.png | Source: ©2026 GabE=mc² — Gabriel Martín del Campo F. & Jazmin Olvera Zacarias. Reproducible from public DK-RD2 code. GitHub: <https://github.com/gabemdelc/DK-RD2-Falsifiable-Predictions> | Zenodo: Zenodo DOI: <https://doi.org/10.5281/zenodo.20637136>

Figure 2 Profiled likelihood constraint on the DK projection factor α_{DK} obtained from DESI BAO and Cosmic Chronometer data. A well-defined minimum is found at $\alpha_{DK}=0.2974$, indicating that the DK correction is observationally constrained rather than arbitrary. The lower panel shows the corresponding profiled sound-horizon scale $r_d(\alpha_{DK})$.

The presence of a localized minimum demonstrates that the DK-RD2 projection factor is not an arbitrary phenomenological parameter. Instead, the observational data constrain the allowed parameter space to a statistically preferred region.

This result is particularly important because it establishes that the thermodynamic–relativistic correction introduced by DK-RD2 behaves as a measurable physical quantity rather than as a freely adjustable degree of freedom. Values significantly different from the best-fit solution produce progressively poorer agreement with the observational datasets. Consequently, future BAO, Cosmic Chronometer, Euclid, and DESI[14] measurements can directly test the stability of the preferred region and determine whether the likelihood minimum persists as observational precision improves.

Together with Figure 1, this analysis provides the statistical foundation for all subsequent DK-RD2 predictions presented in this work.

Figure 3 — Thermodynamic-relativistic coupling structure and cosmological projection

This figure illustrates the thermodynamic-relativistic coupling structure that forms the physical basis of the DK-RD2 framework.

The upper panel compares two gravitational regimes. The first corresponds to the maximal thermodynamic-relativistic coupling,

$$\frac{G_{ab}^{max}}{G_0} = 1 + \frac{v^2 T_0}{c^2 T}$$

representing the fully normalized relativistic limit. The second corresponds to the observable projected coupling,

$$\frac{G_{ab}^{eff}}{G_0} = 1 + \alpha_{DK} \frac{v^2 T_0}{c^2 T}$$

where the projection factor α_{DK} is calibrated directly from DESI[14] BAO and Cosmic Chronometer observations. The shaded region represents the thermodynamic coupling excess that is not projected into the observable gravitational sector. For the globally calibrated value

Figure 03 — DK-RD2 Coupling Structure and Cosmological Projection
 $\alpha_{DK} = 0.2974$ from official global DESI BAO + CC calibration

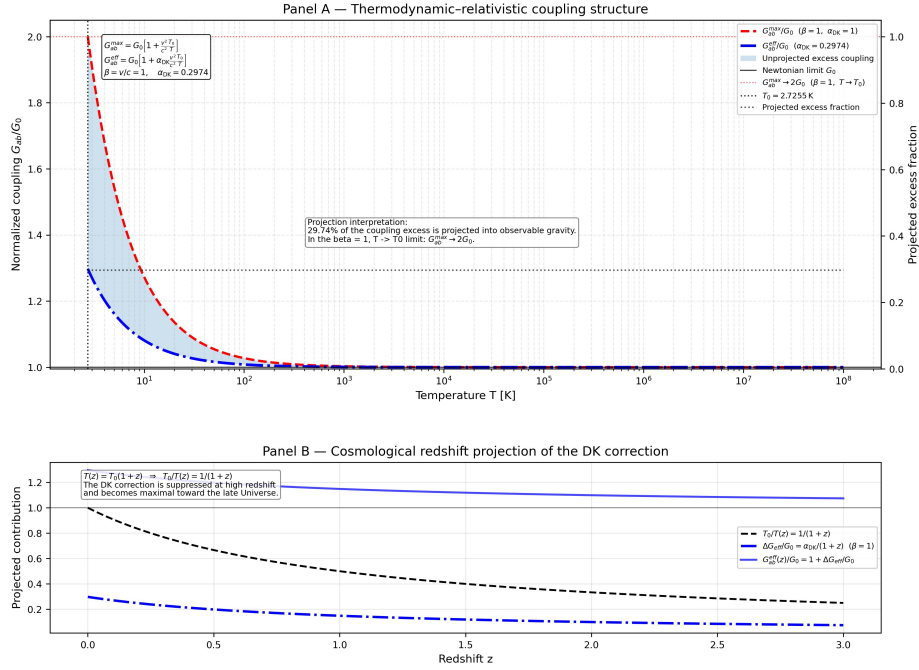


Figure 3 Thermodynamic–relativistic coupling structure of DK-RD2. Panel A shows the maximal and observable projected couplings, highlighting the DESI-calibrated projection factor $\alpha_{DK}=0.2974$ and the bounded saturation limit $G_{ab}^{max} \rightarrow 2G_0$. Panel B illustrates the cosmological suppression of the DK correction with increasing redshift under $T(z) \approx T_0(1+z)$.

The shaded region between both curves represents the thermodynamic coupling excess that is not projected into the observable gravitational sector. For the globally calibrated value

$$\alpha_{DK} = 0.2974$$

approximately 29.74% of the available thermodynamic–relativistic coupling excess contributes to the observable cosmological dynamics. An important feature of the framework is the existence of a bounded relativistic limit. In the regime

$$\beta = \frac{v}{c} \rightarrow 1, T \rightarrow T_0$$

the maximal coupling approaches

$$G_{ab}^{max} \rightarrow 2G_0$$

indicating that the thermodynamic amplification remains finite and saturates dynamically rather than diverging. This behavior prevents runaway gravitational amplification and establishes a natural upper bound for the effective coupling.

The lower panel extends this interpretation to cosmological evolution. Using the standard thermal relation

$$T(z) \approx T_0(1+z)$$

the DK correction becomes

$$\frac{T_0}{T(z)} = \frac{1}{1+z}$$

As a consequence, the observable contribution of the DK coupling decreases naturally with increasing redshift. The correction is strongly suppressed in the early Universe and becomes progressively more relevant as the Universe evolves toward its present thermodynamic state. Within this interpretation, late-time gravitational effects emerge from the thermodynamic projection of the effective gravitational coupling itself, without requiring an explicit dark-energy fluid or cosmological-constant sector.

Together, the two panels establish the physical meaning of the DESI-calibrated projection factor and show how the same thermodynamic–relativistic mechanism simultaneously governs coupling saturation and cosmological evolution. This behavior provides the common physical foundation underlying the expansion-history, effective dark-energy, structure-growth, compact-halo, and Hubble-tension predictions presented in the subsequent sections.

Figure 4 — Effective dark-energy diagnostics reconstructed from the DK-RD2 expansion history

The fourth observational prediction investigates the effective dark-energy behavior that would be inferred by a standard Friedmann–Robertson–Walker (FRW) observer if the underlying cosmological dynamics were governed by the DK-RD2 thermodynamic–relativistic coupling.

Although DK-RD2 does not introduce a fundamental dark-energy fluid or a cosmological constant[6] as an independent physical component, it is possible to reconstruct an effective equation of state $\omega(z)$ from the resulting expansion history. This procedure provides a direct observational comparison with the standard cosmological interpretation used in Λ CDM analyses. The reconstruction is performed through the effective component

$$\Omega_{DE}^{eff}(z) = E^2(z) - \Omega_{m,ref}(1+z)^3$$

where

$$E(z) = \frac{H(z)}{H_0}$$

is the normalized DK-RD2 expansion rate and

$$\Omega_{m,ref} = 0.315$$

corresponds to the Λ CDM reference matter density adopted for comparison. From this quantity, an effective equation of state is reconstructed through

$$\omega(z) = -1 + \frac{1+z}{3} \frac{d \ln \Omega_{DE}^{eff}}{dz}$$

This reconstruction does not imply the existence of a physical dark-energy component within DK-RD2. Instead, it addresses the following diagnostic question:

What effective dark-energy behavior would be inferred if the true cosmological dynamics were generated by the DK-RD2 gravitational coupling?

The upper panel of Figure 4 shows the evolution of the reconstructed component $\Omega_{DE}^{eff}(z)$. Near the present epoch, the DK-RD2 reconstruction remains close to the approximately constant behavior expected in Λ CDM, reproducing a late-time accelerated-expansion regime. At increasing redshift, however, the reconstructed component decreases dynamically rather than remaining constant. The middle panel presents the corresponding effective equation of state $\omega(z)$. At low redshift, the reconstruction remains close to the Λ CDM value

$$\omega = -1$$

indicating that DK-RD2 naturally reproduces an accelerated expansion phase without introducing an explicit cosmological constant[6]. As redshift increases, the reconstructed equation of state departs progressively from the Λ CDM limit, generating an evolving behavior qualitatively similar to the dynamical dark-energy trends suggested by recent DESI analyses.

$$\Delta w(z) = w_{DK}(z) + 1$$

Figure 04 — Explicit Diagnostic Reconstruction of $w(z)$
 $\alpha_{DK} = 0.2974$, $r_d = 149.99$ Mpc (global DESI BAO + CC calibration)

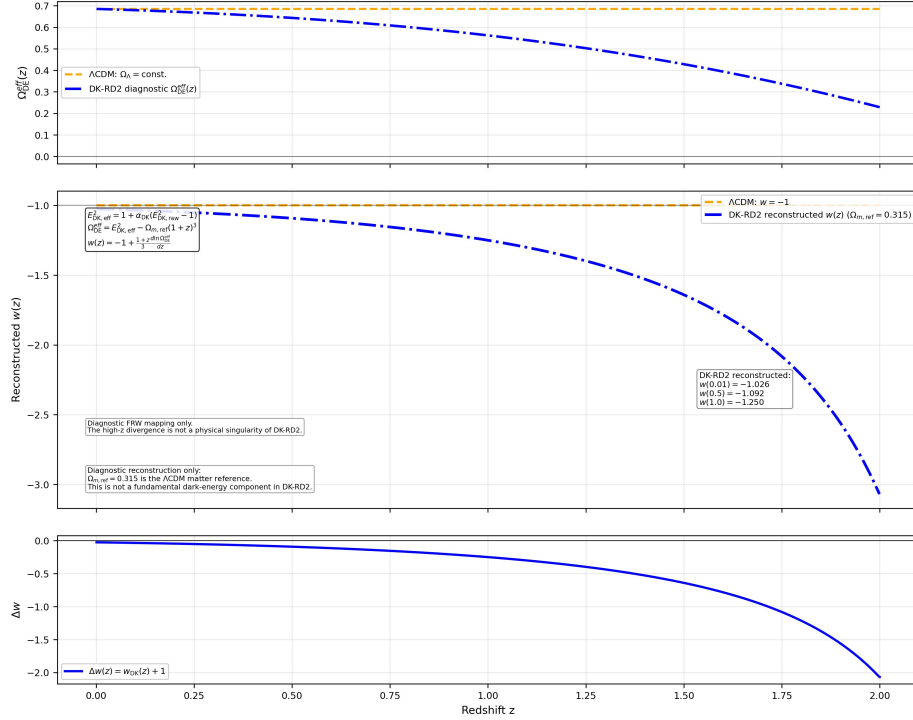


Figure: evidence/fairlyable_Predictions_DK-RD2_image_04.png [Source: ©2024 Galileo+mc — Gabriel Martín del Campo F. & Jaume Olivera Zanarias. Reproducible from public DK-RD2 code. <https://github.com/gamdelcampof/Galileo+mc> Zenodo: Zenodo DOI: <https://doi.org/10.5281/zenodo.2967136>

Figure 4 Diagnostic reconstruction of $\mathcal{Q}_{DE}^{eff}(z)$, the effective equation of state $w(z)$, and the deviation $\Delta w(z) = w_{DK}(z) + 1$ inferred from the DK-RD2 expansion history. The reconstruction reproduces a Λ CDM-like accelerated regime at low redshift while generating an evolving effective dark-energy signature at higher redshift without introducing a fundamental dark-energy component.

highlighting the progressive departure from the Λ CDM cosmological-constant prediction as redshift increases.

At sufficiently high redshift, the reconstructed quantity $\mathcal{Q}_{DE}^{eff}(z)$ approaches zero. Consequently, the logarithmic derivative entering the definition of $w(z)$ becomes singular, producing the apparent divergence visible in the reconstruction. This behavior should not be interpreted as a physical instability of DK-RD2. Rather, it reflects the breakdown of the effective FRW reinterpretation once the inferred dark-energy-like contribution ceases to dominate the expansion history.

It is important to emphasize that $\Omega_{DE}^{eff}(z)$ is a reference-dependent diagnostic quantity constructed using the comparison value $\Omega_{m,ref} = 0.315$. Therefore, its suppression at high redshift reflects the FRW reinterpretation adopted for diagnostic purposes and should not be interpreted as a physical disappearance or instability of the underlying DK-RD2 coupling.

Overall, Figure 4 demonstrates that DK-RD2 naturally reproduces a Λ CDM-like accelerated regime at late times while generating a dynamically evolving effective equation of state at higher redshift without invoking a fundamental cosmological constant[6]. This behavior provides a direct observational discriminator between thermodynamic–relativistic gravitational evolution and conventional dark-energy parameterizations.

Figure 5 — Modified growth histories detectable through large-scale structure surveys

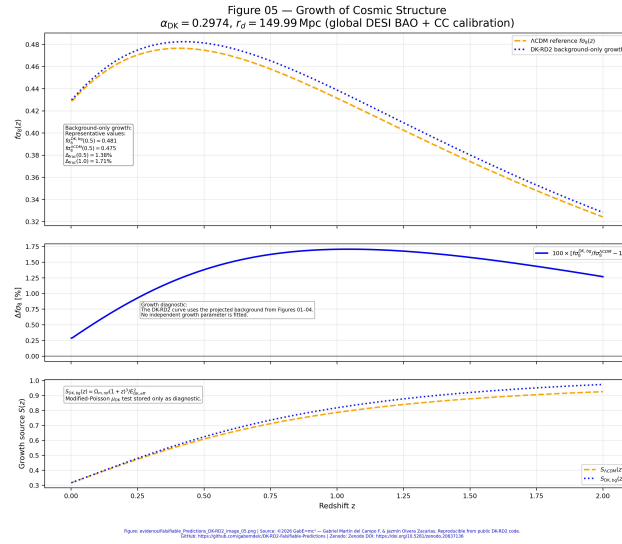


Figure 5 Predicted evolution of $f\sigma_8(z)$ within the DK-RD2 framework. The upper panel compares the DK-RD2 and Λ CDM growth histories, the middle panel shows the fractional deviation, and the lower panel displays the effective growth source term $S(z)$. The prediction follows directly from the DESI BAO + Cosmic Chronometer calibrated thermodynamic–relativistic coupling without additional growth parameters.

The fifth observational prediction examines the growth history of large-scale cosmic structure. In the DK-RD2 framework, the evolution of matter perturbations is computed using the same projected thermodynamic–relativistic background expansion calibrated through DESI Baryon Acoustic Oscillation (BAO) and Cosmic Chronometer (CC) observations.

As a consequence, the linear growth factor and the observable quantity

$$f\sigma_8(z)$$

acquire a characteristic redshift dependence that differs from the standard Λ CDM prediction. Unlike distance-based probes such as supernovae or baryon acoustic oscillations, growth-rate measurements directly test how cosmic structures form and evolve, providing an independent observational channel for evaluating the physical consistency of the framework.

The growth history shown in Figure 5 is obtained from the projected DK-RD2 expansion rate

$$E_{DK,eff}(z) \equiv \frac{H_{DK}(z)}{H_0}$$

without introducing an explicit dark-energy fluid, a modified-Poisson gravitational sector, or an independent phenomenological growth parameter. The resulting evolution therefore represents a direct prediction of the same background dynamics used throughout the previous figures rather than a separate empirical fit.

The reconstructed evolution predicts small but potentially measurable departures from the Λ CDM expectation across the observable redshift range. Because the prediction is derived from the globally calibrated value

$$\alpha_{DK} = 0.2974$$

obtained from the combined DESI BAO and Cosmic Chronometer likelihood, no additional calibration is performed for the growth analysis. The resulting prediction therefore constitutes an independent consistency test of the DK-RD2 framework.

Future measurements from DESI, Euclid, the Vera C. Rubin Observatory, and other large-scale structure surveys will substantially improve observational constraints on

$$f\sigma_8(z)$$

allowing increasingly precise tests of the predicted growth history. Agreement or disagreement with these measurements will therefore provide a direct falsification criterion for the framework.

Together with the expansion-history reconstruction, effective dark-energy diagnostics, compact-halo signatures, and Hubble-tension projections presented in the other figures, the growth of cosmic structure constitutes one of the principal observational probes capable of distinguishing DK-RD2 from Λ CDM and alternative cosmological scenarios.

Figure 6 — Localized gravitational amplification effects in compact galactic and sub-galactic systems

The sixth observational prediction extends the DK-RD2 framework beyond homogeneous cosmology into compact gravitational systems.

Using published core-collapsed Self-Interacting Dark Matter (SIDM) halo reconstructions as an observational benchmark[3], this analysis investigates whether part of the compact-halo phenomenology commonly attributed to highly concentrated dark-matter structures may alternatively emerge from localized thermodynamic–relativistic amplification of the effective gravitational coupling.

Within DK-RD2, highly coherent and thermodynamically organized energy configurations can generate bounded and scale-dependent enhancements of the effective interaction, described by

$$\mu_{eff}(r) = \frac{G_{eff}(r)}{G_0}$$

In this interpretation, the observed gravitational behavior does not necessarily require an increase in unseen mass, but may partially arise from local variations in the effective gravitational coupling. The resulting amplification remains finite and environment-dependent, producing characteristic signatures that can be compared directly with observed compact-halo profiles.

In the standard Λ CDM framework[7, 8], the gravitational coupling remains constant,

$$G_{eff} = G_0$$

and compact-halo phenomena are interpreted primarily through variations in the dark-matter density distribution. DK-RD2 instead predicts an effective amplification of the local gravitational response, providing an alternative observational signature that can be tested independently of the assumed dark-matter profile.

This prediction probes a physical regime fundamentally different from the previous cosmological observables. Whereas Figures 1–5 focus on expansion history, effective dark-energy diagnostics, and structure formation, Figure 6 investigates whether the same thermodynamic–relativistic mechanism leaves measurable signatures in galactic and sub-galactic environments.

The analysis is not intended to replace existing SIDM or Λ CDM interpretations, but rather to evaluate whether part of the observed compact-halo phenomenology can be reproduced through a bounded effective-gravity mechanism. Compact gravitational systems therefore provide an observational test that is largely independent of cosmological expansion measurements.

A robustness analysis was performed by repeating the compact-halo fits over multiple radial fitting intervals. Fornax 6 remained essentially unchanged, while B1938+666 and GD-1 exhibited moderate parameter sensitivity. Nevertheless, the inferred amplification regimes remained qualitatively stable, preserving the distinction between moderate, strong, and extreme amplification scenarios. This indicates that the principal phenomenological conclusions are not driven by the specific fitting interval adopted.

Because the standard Λ CDM expectation corresponds to

$$\mu_{\text{eff}}(r) = 1$$

the predicted amplification profiles provide a direct observational discriminator between a constant-coupling gravitational framework and an effective thermodynamic–relativistic description.

If future observations confirm the predicted amplification signatures, this would suggest that a common thermodynamic–relativistic mechanism may operate across multiple astrophysical scales, linking late-time cosmological evolution with compact gravitational phenomena. Conversely, the absence of such signatures would place direct constraints on the applicability of DK-RD2 in highly concentrated gravitational environments.

Together with the previous figures, this prediction extends the DK-RD2 falsification program from cosmological observables to galactic and sub-galactic systems, providing a genuinely multi-scale test of the framework.

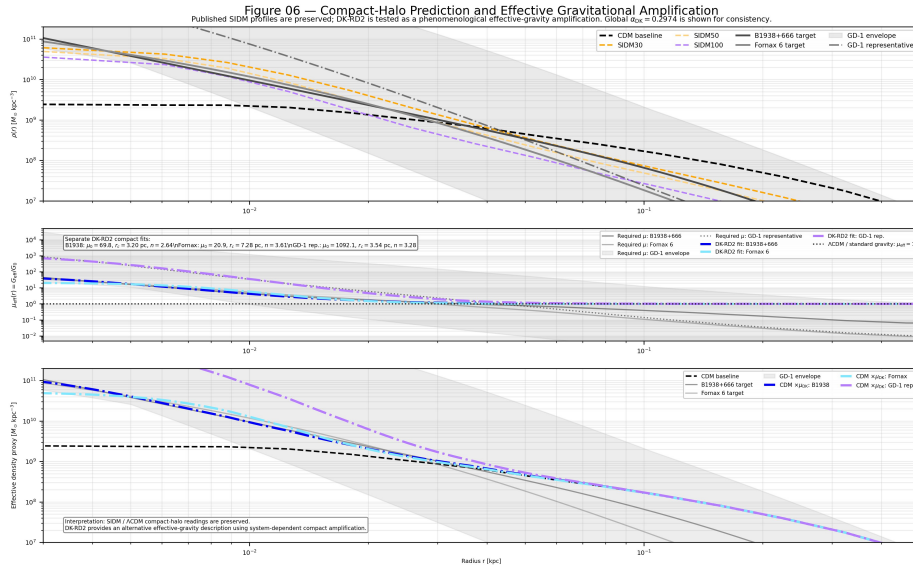


Figure: evidence/falsifiable_Predictions_DK-RD2_image_06.png | Source: 130220 Dark-matter — Gabriel Martínez del Campo, F. & Javier Ojeda-Davila. Reproducible from public DK-RD2 code. [Github: https://github.com/gdmartinez/DK-RD2-falsifiable-Predictions](https://github.com/gdmartinez/DK-RD2-falsifiable-Predictions) | Zenodo: Zenodo DOI: <https://doi.org/10.5281/zenodo.29637126> | External companion data: Zenodo: Mathematics notebook from <https://doi.org/10.5281/zenodo.2912409>

Figure 6 Compact-halo prediction and effective gravitational amplification in DK-RD2. The upper panel shows representative compact-halo density profiles and SIDM benchmarks. The middle panel presents the inferred amplification profiles, $\mu_{eff}(r) = G_{eff}(r)/G_0$ while the lower panel shows the corresponding effective density profiles obtained by applying the DK-RD2 amplification to the CDM baseline. Robustness tests performed over multiple fitting intervals indicate that the inferred amplification regimes remain qualitatively stable despite moderate parameter variations.

Find the result robustness test file in “evidence/Falsifiable_Predictions_DK-RD2_table_06_robustness.csv”, using published SIDM halo reconstructions and their associated public Zenodo notebook/data release [16].

Figure 7 — Thermodynamic projection mechanism for distinct H_0 reconstructions

The seventh observational prediction investigates whether thermodynamic evolution can generate distinct effective reconstructions of the Hubble constant between early- and late-universe observations.

Within DK-RD2, the effective gravitational coupling evolves according to

$$G_{ab}(T, \nu) \approx G_0 \left(1 + \alpha_{DK} \frac{\nu^2 T_0}{c^2 T} \right)$$

where the thermodynamic factor

$$\frac{T_0}{T(z)}$$

naturally suppresses the DK correction at high redshift while progressively enhancing its observable contribution as the Universe approaches the present cosmological epoch. For cosmological applications,

$$T(z) \approx T_0(1 + z)$$

implying that observations probing the early Universe and those probing the late Universe may effectively sample different thermodynamic–relativistic gravitational regimes, even when describing the same underlying expansion history.

As a consequence, the framework predicts that cosmological inferences based on early-Universe observables, such as the Cosmic Microwave Background, and those based on late-Universe observables, such as Cepheid-calibrated Type Ia supernovae [4,5], may yield systematically different effective reconstructions of the Hubble constant

$$H_0$$

without requiring additional cosmological sectors, exotic dark-energy components, or modifications external to the DK-RD2 framework.

This prediction is particularly significant because it provides a direct and falsifiable mechanism capable of generating systematic differences between early- and late-Universe determinations of the Hubble constant. Rather than introducing new forms of

matter or energy, DK-RD2 attributes part of the discrepancy to the thermodynamic projection of the effective gravitational coupling calibrated through the DESI BAO + Cosmic Chronometer observations.

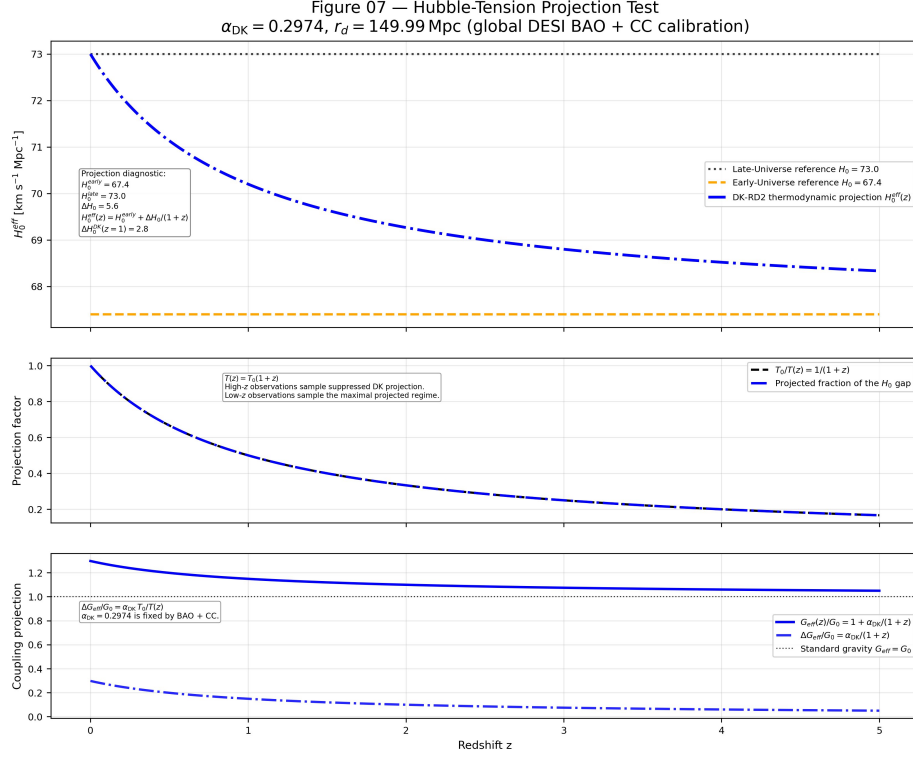


Figure 7 DK-RD2 thermodynamic projection of the Hubble constant. The figure shows how thermodynamic-relativistic gravitational evolution can generate distinct effective reconstructions of H_0 between early and late Universe observations, providing a potential explanation for the Hubble tension.

The selected H_0 values are representative of the currently reported early-late-Universe determinations and are used only to illustrate the DK-RD2 thermodynamic projection mechanism.

Figure 7 provides a quantitative realization of this hypothesis. The upper panel shows the effective reconstruction

$$H_0^{\text{eff}}(z)$$

obtained from the globally calibrated projection factor

$$\alpha_{DK} = 0.2974$$

and compared with representative early and late Universe reference measurements.

The middle panel displays the thermal suppression factor

$$\frac{T_0}{T(z)} = \frac{1}{1+z}$$

which governs the observable contribution of the DK correction throughout cosmic history. The lower panel presents the corresponding evolution of the effective gravitational coupling,

$$\frac{G_{eff}(z)}{G_0}$$

together with the projected excess contribution

$$\frac{\Delta G_{eff}(z)}{G_0}$$

Taken together, these diagnostics illustrate how a single thermodynamic–relativistic coupling can generate distinct effective reconstructions of H_0 without introducing an explicit dark-energy sector.

The figure therefore defines a direct observational test. If future measurements demonstrate that early- and late-Universe determinations converge toward a single thermodynamic-independent value with increasing precision, the DK-RD2 interpretation would be disfavored. Conversely, the persistence of systematic differences correlated with the effective thermodynamic regime probed by each observational method would support the thermodynamic–relativistic interpretation.

Together with the coupling structure of Figure 3, the effective dark-energy reconstruction of Figure 4, the growth-history prediction of Figure 5, and the compact-halo analysis of Figure 6, this prediction extends the DK-RD2 falsification program to independent determinations of the Hubble constant.

Figure 7 therefore represents one of the most direct observational discriminators of the framework. Future measurements from DESI, Euclid, the Vera C. Rubin Observatory, the Nancy Grace Roman Space Telescope, and related observational programs will provide increasingly stringent tests of whether thermodynamic–relativistic gravitational evolution can generate measurable differences between early and late Universe cosmological inferences.

Figure 8 — Observational discrimination between DK-RD2 and alternative explanations of recent DESI observations

The final analysis examines whether recent DESI-scale deviations can be explained by fundamentally different physical mechanisms and, more importantly, whether future observations can discriminate among those competing interpretations[13].

Several explanations have been proposed to account for possible departures from the standard Λ CDM framework[13]. Among the most actively discussed are evolving dark-energy models, Hot New Early Dark Energy (Hot NEDE[11]), and Dark Acoustic Oscillation (DAO) scenarios associated with Dark Radiation–Matter Decoupling (DRMD). Although these frameworks may reproduce similar large-scale observational trends, they generally require the introduction of additional physical sectors, including evolving dark-energy fluids, dark radiation, early-energy injections, phase transitions, or interactions within the dark sector.

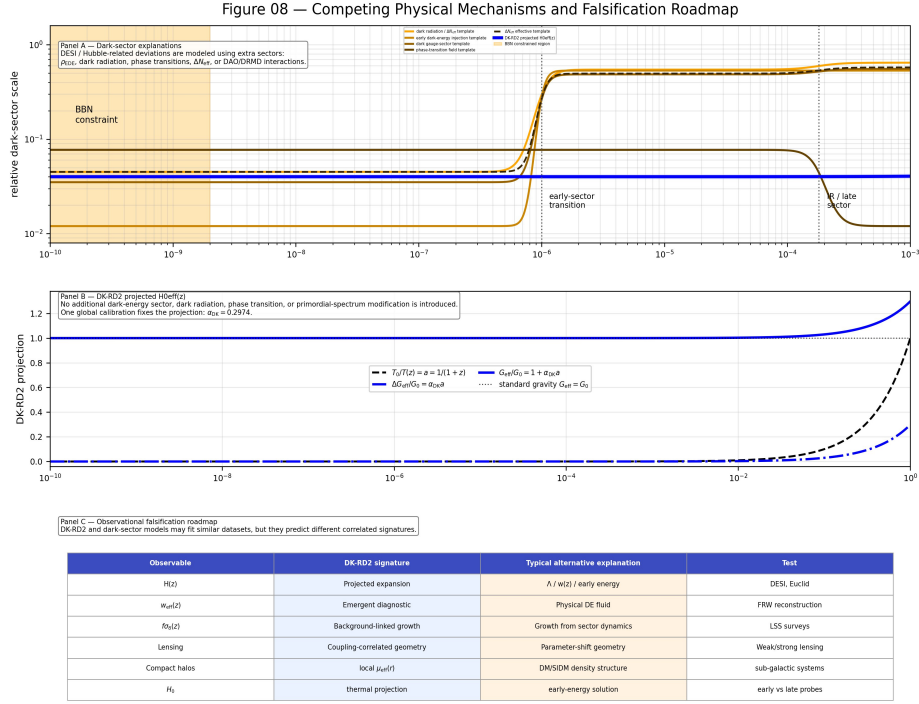


Figure: evidence/falsifiable_predictions_DKRD2_image_08.png | Source: ©2020 Gold+cosmic — Gabriel Murthy del Campo F. & Jazmin Olvera Zucatlán. Reproducible from public DK-RD2 code.
 GitHub: <https://github.com/goldcosmic/DKRD2-falsifiable-Predictions> | Zenodo: <https://doi.org/10.5281/zenodo.2803718>

Figure 8 Observational discrimination between DK-RD2 and alternative explanations of recent DESI observations. Panel A illustrates representative dark-sector extensions commonly invoked to explain DESI-scale deviations. Panel B shows the DK-RD2 interpretation based on thermodynamic evolution of the effective gravitational coupling. Panel C highlights the principal observational domains where future measurements can distinguish between these competing physical mechanisms.

In evolving dark-energy models, the observed behavior is interpreted as evidence for a redshift-dependent equation of state,

$$w(z) \neq -1$$

while Hot NEDE[11] models introduce a transient early-energy component that modifies the expansion history and leaves characteristic signatures in Cosmic Microwave Background observables. DAO/DRMD[12] scenarios invoke interactions between dark matter and dark radiation, generating distinctive signatures in the matter power spectrum and the evolution of cosmic structure.

In contrast, DK-RD2 attributes the observed deviations to the thermodynamic evolution of the effective gravitational interaction itself. Within this framework, the apparent departures from Λ CDM emerge from the evolution of the effective coupling with the thermal state of the Universe, without introducing additional dark-energy fluids, dark-radiation sectors, early-energy injections, phase transitions, or modifications of the primordial power spectrum.

Because the underlying physical mechanisms differ, the competing frameworks predict different correlations among cosmological observables. These differences extend beyond expansion-history measurements and affect effective dark-energy reconstructions, structure growth, gravitational lensing, compact gravitational systems, and independent determinations of the Hubble constant.

Figure 8 therefore focuses on the physical origin of the proposed explanations rather than on parameter fitting alone. Panel A illustrates representative classes of additional dark-sector physics commonly invoked to explain DESI-scale deviations. Panel B presents the DK-RD2 interpretation, in which the same observational phenomena emerge from the thermodynamic evolution of the effective gravitational coupling. Panel C summarizes the principal observational domains where future measurements can discriminate between these competing physical mechanisms.

This comparison represents the final step of the DK-RD2 falsification program. Whereas Figures 01–07 establish a sequence of internally consistent and observationally testable predictions, Figure 8 places those predictions within the broader context of contemporary cosmology and identifies the measurements most capable of distinguishing between competing explanations of recent DESI observations.

If future observations converge toward the correlated signatures predicted by DK-RD2 across expansion history, effective dark-energy diagnostics, structure growth, gravitational lensing, compact halos, and Hubble-constant determinations, the thermodynamic–relativistic interpretation would gain empirical support. Conversely, if the data favor the signatures expected from additional dark-sector physics, DK-RD2 would be directly challenged.

In this sense, Figure 8 shifts the discussion from cosmological fitting toward experimental discrimination between competing physical mechanisms. The central question

is no longer whether multiple frameworks can reproduce current observations, but whether future observations can determine if the observed phenomena arise from additional dark-sector components or from the thermodynamic evolution of gravity itself.

5 Discussion and Conclusions

An important outcome of this work is that the DK-RD2 framework provides a fundamentally different strategy for interpreting recent cosmological observations.

Many contemporary explanations of DESI-scale deviations introduce additional physical sectors, including evolving dark-energy fluids, early-dark-energy components, dark-radiation interactions, phase transitions, or modifications of the primordial perturbation spectrum. Although these approaches may successfully reproduce specific observational datasets, they generally do so by extending the particle or energy content of the cosmological model.

DK-RD2 follows a different path. Rather than introducing additional sectors, the framework attributes the observed phenomena to the thermodynamic evolution of the effective gravitational interaction itself. In this interpretation, cosmic acceleration, effective dark-energy behavior, Hubble-tension projections, structure-growth modifications, and compact gravitational amplification emerge as different manifestations of a common thermodynamic–relativistic mechanism operating across multiple physical scales.

The significance of this distinction is not that DK-RD2 eliminates the need for further theoretical development. On the contrary, several aspects remain open and require deeper investigation. In particular, the development of a fully covariant formulation and a perturbative treatment derived from first principles constitute important directions for future work. However, the framework demonstrates that a broad class of observational phenomena can be described without invoking new dark-energy fluids, dark-radiation sectors, or early-energy injections, thereby providing a physically distinct and directly testable alternative to conventional dark-sector explanations.

The primary contribution of the present work is therefore not the optimization of cosmological fits, but the construction of a falsifiable observational program. Figures 1–8 identify a set of correlated signatures spanning expansion history, effective dark-energy diagnostics, structure growth, gravitational lensing, compact halos, and independent determinations of the Hubble constant. Future observations can test whether these signatures emerge as a coherent pattern, thereby distinguishing between thermodynamic–relativistic gravitational evolution and competing cosmological interpretations.

Ultimately, DK-RD2 proposes that gravity may not be a primitive interaction with fixed strength, but rather an emergent manifestation of how energy organizes, stabilizes, and couples to spacetime. If this interpretation is correct, spacetime geometry, gravitational interaction, cosmological evolution, and compact gravitational phenom-

ena may represent different expressions of a single thermodynamic process acting across the observable Universe.

Beyond the specific predictions presented here, the framework defines a broader observational roadmap. Its central prediction is not an isolated anomaly in a single dataset, but a correlated set of signatures appearing across multiple independent observables. The decisive test of the framework will therefore not be whether any individual prediction is confirmed, but whether the complete observational pattern emerges consistently across cosmological expansion, effective dark-energy reconstructions, structure formation, gravitational lensing, compact gravitational systems, and Hubble-constant determinations.

Future observational campaigns may therefore evaluate not only the individual predictions of DK-RD2, but also the internal consistency of the thermodynamic–relativistic interpretation itself. The development of dedicated observational strategies capable of discriminating between thermodynamic-gravity scenarios and dark-sector explanations represents a natural extension of the present work and will be explored in future studies.

References

- [1] Martín del Campo Flores, G. (2026). DK-RD2: Duo-Kinetic Relativistic Dynamics 2, Thermodynamic–Relativistic Gravity and the Origin of Cosmic Expansion. <https://doi.org/10.5281/zenodo.18529901>
- [2] Martín del Campo Flores, G. (2026). What Is Gravity? From Emergent Mass to Thermodynamic–Geometric Dynamics in the DK-RD2 Framework <https://doi.org/10.5281/zenodo.20078175>
- [3] Yu, H.-B. (2026). Three Birds with One Stone: Core-Collapsed SIDM Halos as the Common Origin of Dense Perturbations in Lenses, Streams, and Satellites. arXiv:2510.11006 [astro-ph.GA].
- [4] Riess, A.G. et al. Observational evidence from supernovae for an accelerating universe and a cosmological constant. *Astron. J.* 1998, 116(3), 1009.
- [5] Perlmutter, S. et al. Measurements of Ω and Λ from 42 high-redshift supernovae. *Astrophys. J.* 1999, 517(2), 565.
- [6] Weinberg, S. The cosmological constant problem. *Rev. Mod. Phys.* 1989, 61(1).
- [7] Liddle, A.R. *An Introduction to Modern Cosmology*, 3rd ed.; Wiley: Hoboken, NJ, USA, 2015.
- [8] Mukhanov, V. *Physical Foundations of Cosmology*; Cambridge University Press: Cambridge, UK, 2005.
- [9] Verlinde, E.P. On the Origin of Gravity and the Laws of Newton. *J. High Energy Phys.* 2011, 2011(4), 29.
- [10] Jacobson, T. (1995). *Thermodynamics of spacetime: The Einstein equation of state*. *Phys. Rev. Lett.* **75**, 1260–1263.

- [11] Niedermann et al.
Hot New Early Dark Energy.
Phys. Rev. D 105, 063509 – Published 8 March, 2022
DOI: <https://doi.org/10.1103/PhysRevD.105.063509>
- [12] Buckley, M. R., Cyr-Racine, F.-Y., Sigurdson, K., & Zavala, J. (2014).
Scattering, Damping, and Acoustic Oscillations: Simulating the Structure of
Dark Matter Interacting with Dark Radiation.
Physical Review D, 90, 043524. DOI: 10.1103/PhysRevD.90.043524
- [13] Camarena, D.; Greene, K.; Houghteling, J.; Cyr-Racine, F.-Y. (2025).
DESIGNing Concordant Distances in the Age of Precision Cosmology: The Im-
pact of Density Fluctuations. arXiv:2507.17969

Dataset Sources

The BAO datasets employed throughout this work correspond to the DESI DR2 processed likelihood compilation distributed through the CobayaSampler public repository, including the combined Gaussian BAO measurements and covariance matrices used in the statistical calibration of α_{DK} and r_d .

- [14] Adame, A.G. et al. (DESI Collaboration) (2025).
DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon
Acoustic Oscillations. *Journal of Cosmology and Astroparticle Physics*,
2025(02), 021. <https://doi.org/10.1088/1475-7516/2025/02/02>
- [15] CobayaSampler Collaboration (2026).
DESI BAO DR2 likelihood data repository. GitHub repository. Available at:
[CobayaSampler BAO Data Repository](#)
- [16] Yu, H.-B. (2026). *Three Birds with One Stone: Core-Collapsed SIDM Halos
as the Common Origin of Dense Perturbations in Lenses, Streams, and Satellites*.
arXiv:2510.11006. Associated data/notebook: Zenodo.
<https://doi.org/10.5281/zenodo.19116269>

Conflicts of Interest: Conflicts of Interest, Data Access, Ethics, and Financial Disclosure Statement

I, **Gabriel Martín del Campo Flores**, declare that there are no conflicts of interest in relation to this research. I am an independent researcher and do not receive funding, support, or donations from any public or private entity that may influence the results or conclusions of this study. Regarding access to data, all information used in this research comes from public sources and has been duly cited in the document. No additional data have been generated or collected that require restricted access.

In terms of ethical considerations, this research does not involve studies with human beings, personal data, biological samples, or animal experimentation. Therefore, the approval of an ethics committee was not required.

In the development and preparation of this manuscript, I made use of artificial intelligence tools to assist in the structuring and refinement of the language. However, all

scientific ideas, physical models, theoretical developments, numerical implementations, and conceptual contributions are entirely original and of my own authorship. No part of this work is a copy or adaptation of external theories not properly credited.

The author also acknowledges the collaboration of **Jazmín Olvera Zacarías** in scientific communication, editorial review, and manuscript presentation. Her contribution focused on improving clarity, readability, and dissemination of the work. She did not participate in the development of the theoretical framework, physical models, mathematical derivations, numerical calculations, statistical analyses, or scientific conclusions presented in this study. Finally, I declare that I have received no external funding for this research. All associated costs have been independently covered by the author.

Jazmín Olvera Zacarias
ORCID: 0009-0007-1771-4076
E-mail: Jaolveraz@ipn.mx

Gabriel Martín del Campo Flores
ORCID: 0009-0009-7106-9519
Email: gabemdelc@gmail.com

GitHub <https://github.com/gabemdelc/DK-RD2-Falsifiable-Predictions>

This work builds upon the DK-RD2 framework previously introduced in:
Martín del Campo Flores, G. (2026).
DK-RD2: Duo-Kinetic Relativistic Dynamics 2, Thermodynamic–Relativistic
Gravity and the Origin of Cosmic Expansion.
Zenodo. DOI: <https://doi.org/10.5281/zenodo.20078175>

[El Salto, Guadalajara, Jalisco, Mexico, June/2026]